

Constants as the bridge

Minimal-shot autonomy, in motion

Snow-Underlay · SoTA Commission I · May 2026

A snow plough's job is short.

Keep the road clear.

The catch:

**while the plough is doing it,
the road is invisible.**

A self-driving stack trained on
Cityscapes
will report, with calibrated
confidence,
that the entire scene is sky.

Off-the-shelf segmentation, applied directly:

a confident red mask
that does not survive a glance.

Or worse — silent failure:
zero road predicted,
no warning issued.

*27 million miles of road.
The long tail of conditions
any of them can be in
is longer than the road itself.*

We are not going to label our way out of it.

There is a different move.

For almost every regime where autonomy fails for lack of data, there is an *adjacent regime* — temporally, seasonally, geographically — where data exists, and where the parts that matter are the same.

The plough's road is the same road it was last July.

The curb hasn't moved. The hydrant hasn't moved.

The road's *appearance* has changed completely.

The road's **position in space** has not.

Constants as the bridge

If we can identify what stays
constant
between the data-rich regime
and the data-poor one,
we can extend our existing
models into the new regime —
**without learning a single
thing about it.**

The recipe (per snow frame)

1. Pull the **live snowy frame** from the plough's camera.
2. Pull a **clear-season prior** of the same coordinates (Boreas paired summer drives).
3. **Match** the two with a frozen feature matcher.
4. Estimate a **homography**, biased toward the ground plane.
5. Run a Cityscapes road segmenter — **on the clear prior only**.
6. **Warp** the road mask onto the snowy frame.

The plough now knows where the road is.

It has not been trained on snow.

Architecture

Component	Role	Pretrained on
DISK (<i>NeurIPS '20</i>)	Local features	MegaDepth · no snow
LightGlue (<i>ICCV '23</i>)	Sparse matcher	MegaDepth · no snow
USAC-MAGSAC (<i>CVPR '20</i>)	Robust homography	—
Mask2Former (<i>CVPR '22</i>)	Road segmenter	Cityscapes · no snow
Boreas (<i>IJRR '23</i>)	Snow + paired summer captures	— (CC BY 4.0)

Every learned component is **frozen**.
Snow appears only at inference, as the runtime input.

In motion

A 15-second snow drive on a buried Toronto residential street.

Per snow frame: $K=3$ nearest summer priors, snow $\rightarrow$ summer match, warp the segmenter's road mask back, fuse, EMA-smooth.

The cross-season principle survives motion.

`outputs/video/boreas_2021_01_26/overlay.mp4` — 15 s, 1024×856, ~7 MB

What we tried that didn't work

A pair with 3–7× more inliers can still produce a worse video.

Synthetic priors from past snow frames matched dramatically better in stills.

In motion, each frame's slightly-too-large mask seeded the next frame's prior; the road overlay drifted outward into bushes within seconds. *Positive feedback loop.*

Optical-flow propagation between matched keyframes: vanishing-point flow stretches the previous mask outward at every step. Same outcome, different mechanism.

EMA on the binary mask, $\alpha = 0.4$, was what survived the motion test. Failure modes are evidence too.

Minimal-shot integrity

Claim	Status
Zero snowy frames touch any model weights	✓
Zero snowy frames touch any annotation pipeline	✓
Snow appears only as runtime input	✓
Pretrained matcher · pretrained segmenter · classical RANSAC	✓
Reproducible from a clean clone with one command (<code>make reproduce</code>)	✓

The only handle we offered the model on the snow regime was the clear prior of the same place.

Generalising

The structure of the move:

A model trained on regime A.

An inference target in regime B.

A known correspondence between the two.

Transfer through the correspondence.

Snow on a road is one instance.

Low-light medical imaging without low-light training data.

Polar earth observation without polar training data.

A manipulator on Mars without Mars training data.

Each admits the same structure.

Find what stays the same

and walk across.

Constants as the bridge.

Reproduce

```
git clone <repo>
cd snow-underlay
uv sync --python 3.12
make reproduce
```

Read

[README.md](#) · [docs/writeup.pdf](#) · [notebooks/02_video_walkthrough.ipynb](#)

Submission

SoTA Commission I — Minimal-Shot Autonomy · May 2026 · Boreas dataset CC BY 4.0

Appendix · submission-video plan

- Title: Constants as the bridge — in motion.*
- Length: 90 s (target). Acceptable range 60–120 s.*
- Format: 1080p MP4, H.264 / yuv420p, ≤ 50 MB.*
- Composed externally (Premiere / Final Cut / DaVinci / etc.); this section is the editor's brief.*

Asset inventory

Every asset below is produced by a `make` command. No one-offs.

Moving overlay clips (15 s each, 1024 × 856)

Asset	Path	Produced by
Canonical overlay (snow + green road, headline)	outputs/video/boreas_2021_01_26/overlay.mp4	make reproduce